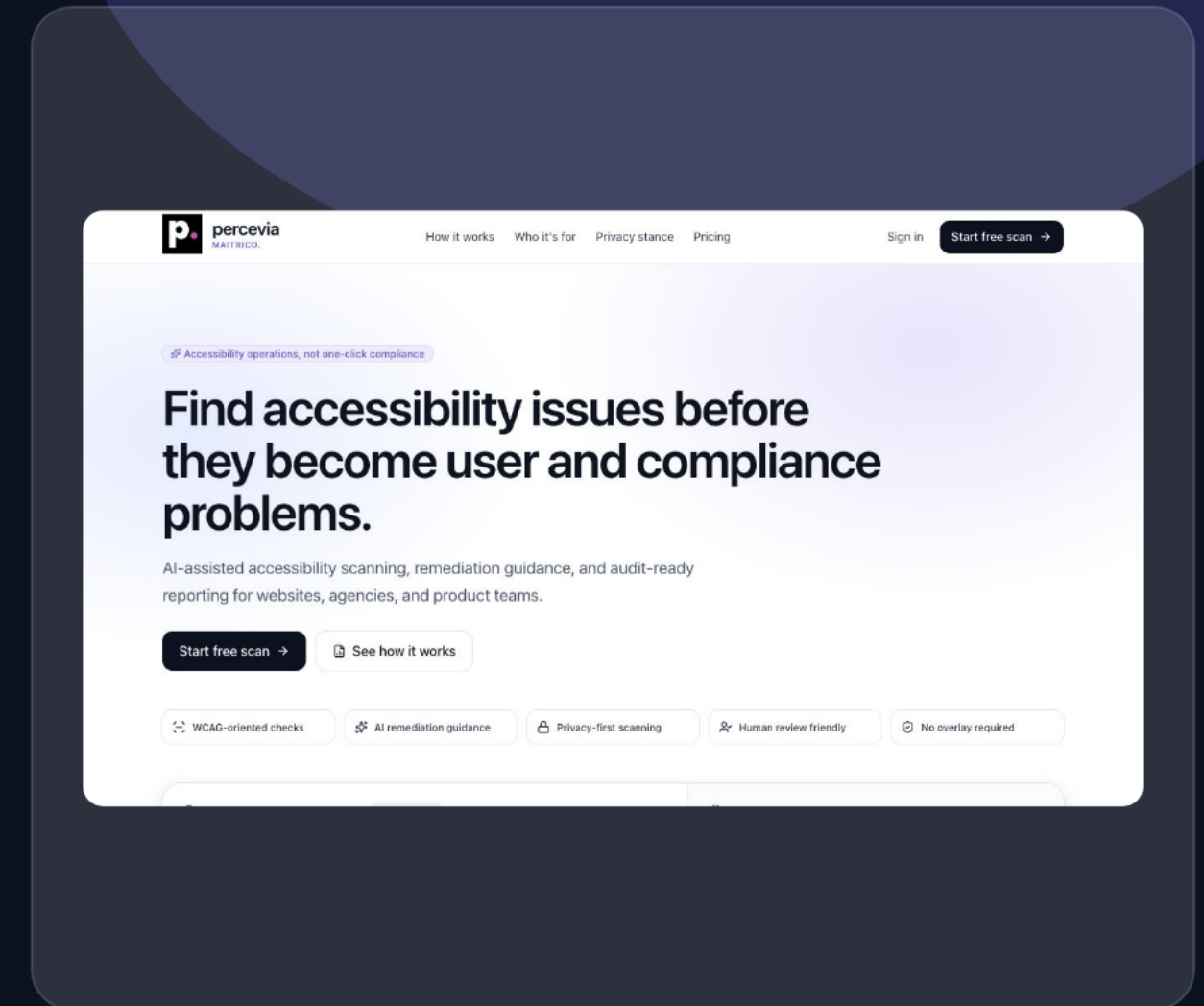


Percevia AI

Project scope and technical architecture presentation

Prepared from the English technical report
Review date: June 23, 2026



Executive summary

Percevia AI turns web accessibility from a one-time audit into a repeatable operational process.

- Scans real pages with Playwright, Chromium, and axe-core.
- Tests desktop, tablet, mobile, and key interactive states.
- Groups repeated defects by likely root cause.
- Creates remediation tasks, reports, comparisons, and monitor schedules.
- Keeps AI and visual evidence opt-in, bounded, and auditable.
- Avoids certification language and does not guarantee legal compliance.

33

analysis variants

Max per page.

323

passing tests

Unit and integration checks.

4

export formats

HTML, PDF, CSV,
and JSON.

5

plan tiers

Free through
Enterprise.

The problem Percevia AI solves

01 Tools produce noise

Accessibility scanners often return hundreds of repeated findings without explaining shared root causes.

02 Teams lack priority

Developers, PMs, and auditors need severity, risk, affected pages, and owner-ready tasks.

03 Reports are fragmented

Executives need a clear risk story while developers need technical evidence, WCAG tags, and selectors.

04 Audits do not stick

One-time audits miss regressions unless teams can re-scan, compare, and monitor over time.

05 Privacy is unclear

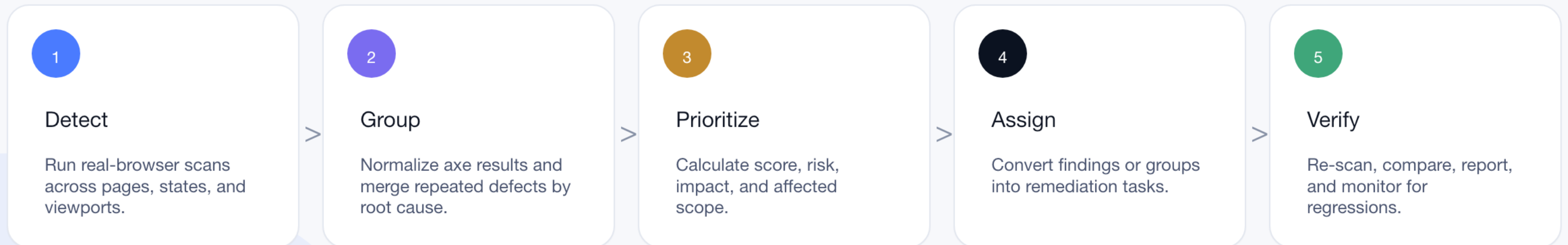
Screenshots, HTML samples, AI processing, and retention must be controlled with explicit consent.

06 Compliance claims are risky

Automated results must support decisions without implying certification or legal guarantees.

Product approach

Percevia AI is designed as an accessibility operations loop, not just a scanner.



Operating principle: automated analysis is decision support. Human review remains part of the accessibility process.

Capability map

Scanning

Single, multi-page, sitemap, and manual URL list modes

Operations

Statuses, root-cause groups, remediation tasks, comparison

Evidence

Opt-in screenshots, selector metadata, redaction, expiry

Monitoring

Plan-based recurring scans with stored thresholds.

Analysis

axe-core normalization, WCAG tags, severity, review flags

AI

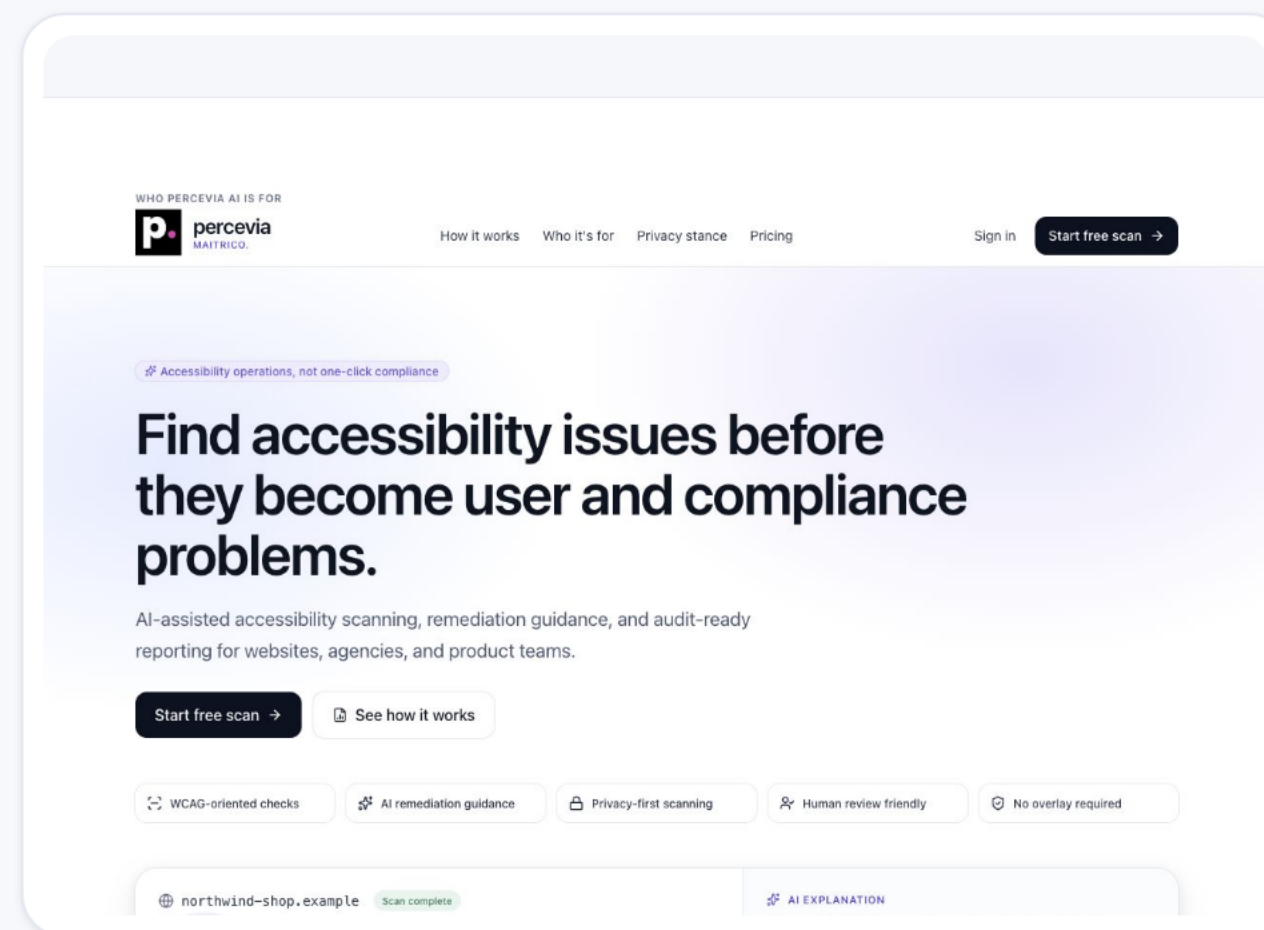
Opt-in explanations, code suggestions, and project assistant

Reporting

Executive, technical, CSV, PDF, JSON, and public links

Governance

Roles, privacy center, audit logs, export, deletion.



Audience and role model

01 Product managers

Visibility into risk, scope, roadmap, and progress.

02 Developers

Selectors, failing HTML, WCAG tags, code guidance, tasks.

03 Auditors

Review flags, false-positive states, manual checklist support.

04 Agencies

Multi-client reporting, client viewers, branded exports.

05 Executives

Readable risk summaries without low-level technical noise.

06 Privacy teams

AI consent, screenshots, retention, deletion, audit logs.

Roles include owner, admin, developer, auditor, client_viewer, and report_viewer.

Permissions separately control scan creation, AI access, report export, billing, privacy, team management, deletion, and remediation.

Product scope and current status



Implemented

- Auth and onboarding
- Scan creation and results
- Root-cause grouping
- Tasks and statuses
- HTML/CSV/JSON reports
- Privacy center and deletion
- Team invitations



Configuration-dependent

- OpenAI explanations
- Visual evidence storage
- Polar billing
- PDF report generation
- White-label public reports



Incomplete

- Monitor email delivery
- Slack alert delivery
- Some stale UI or docs copy
- Lint quality gate

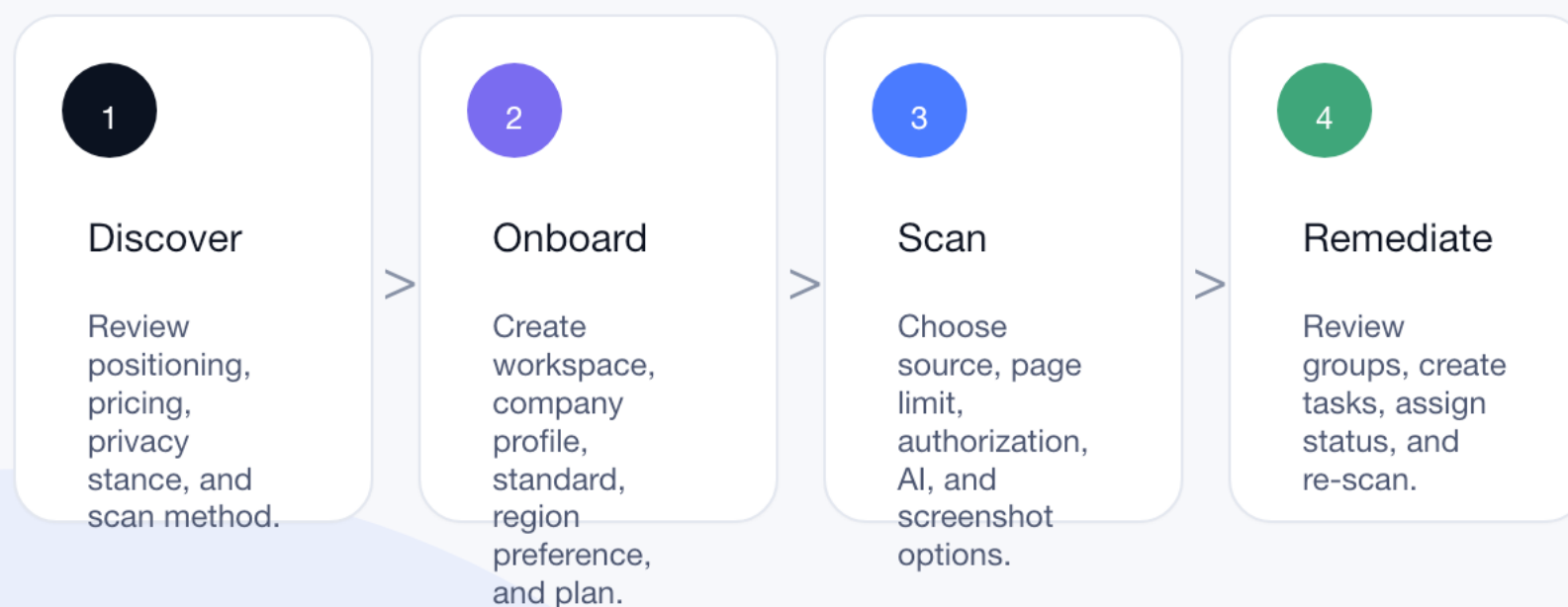


Roadmap

- Enterprise SSO
- Private scanner agent
- Dedicated regions
- Large inventory
- Jira/Linear/GitHub integrations

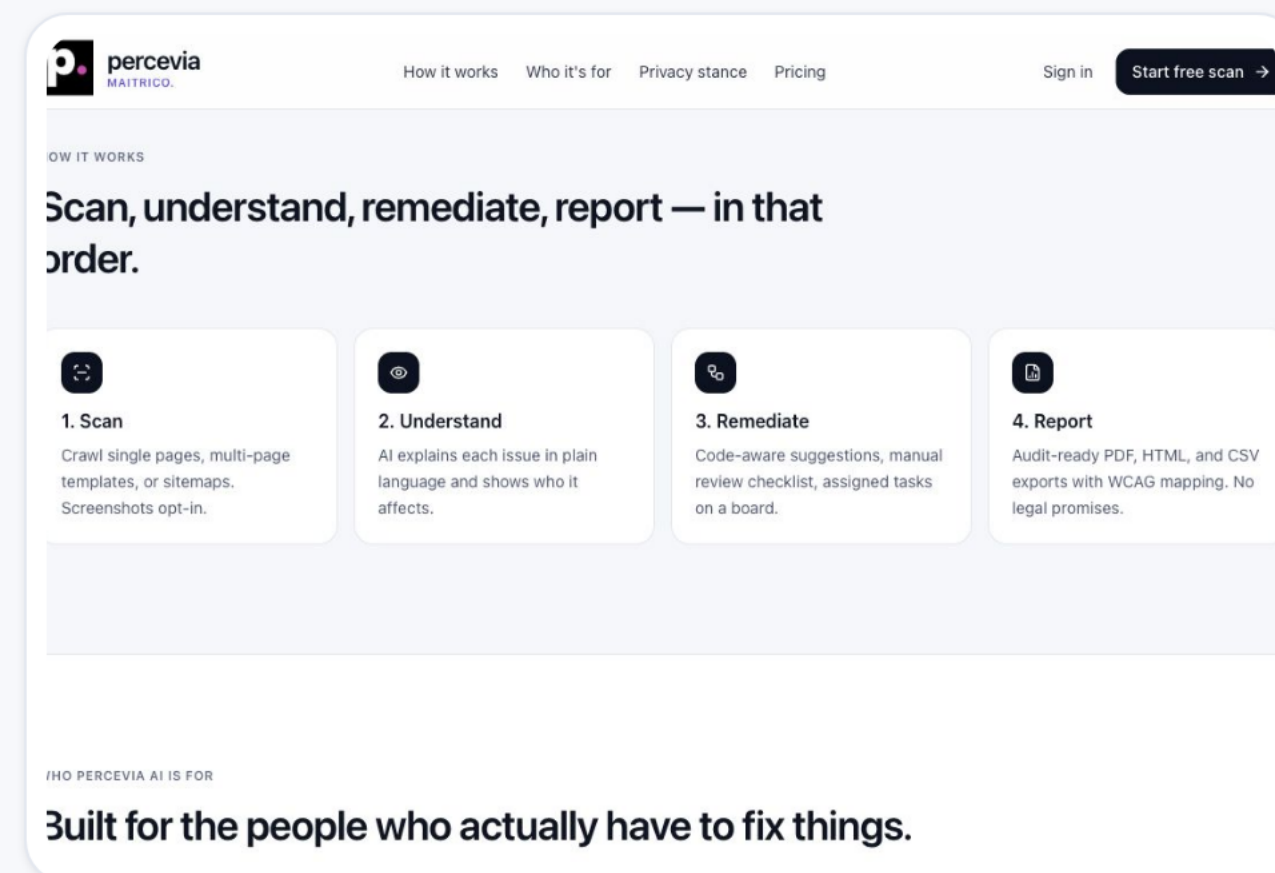
The product is feature-rich today, but several commercial and operational features depend on production configuration or near-term follow-through.

End-to-end user journey

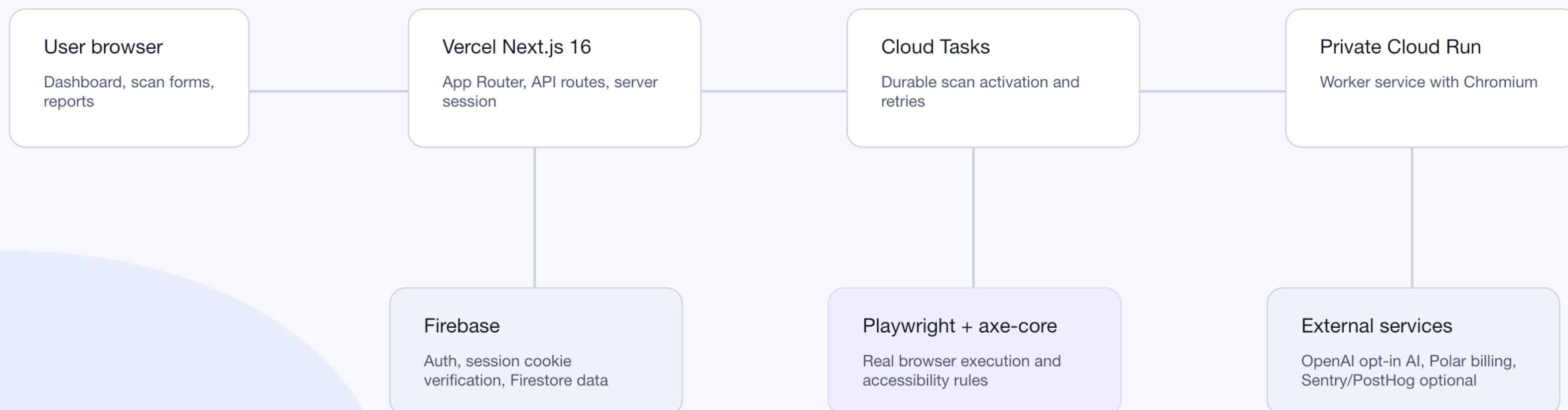


Reporting loop

Completed scans generate executive, technical, CSV-oriented, and public-shareable outputs. The same scan history supports future comparison and monitoring.



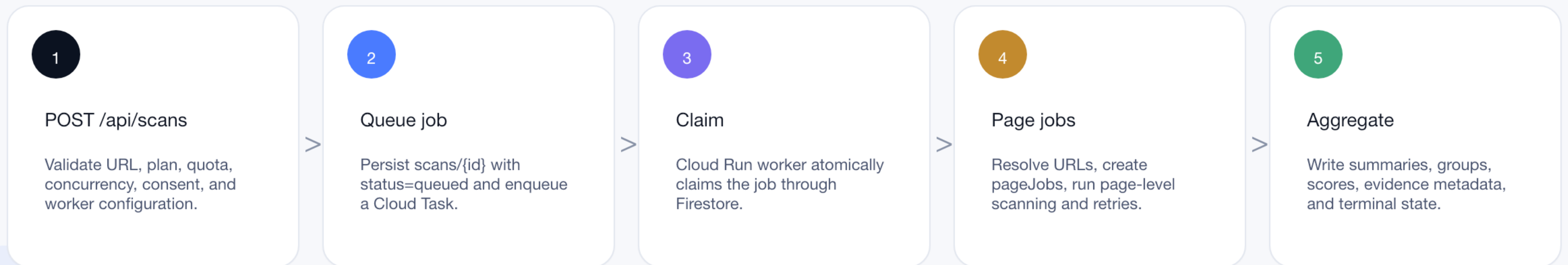
Current system architecture



Production stance

Scan creation does not run Chromium directly. It writes a queued job; Cloud Tasks wakes a private Cloud Run worker to claim, scan, persist, and aggregate results.

Scan processing pipeline



Scan phases

crawling, scanning, aggregating, completed, completed_with_errors, failed

Page-job states

queued, running, completed, failed. One slow page does not block the full scan.

Recovery behavior

Heartbeats, stale detection, queue timeout, sweeper, requeue, and aggregation restart.

Scan engine technical model

4

scan sources

single, multi-page, sitemap,
manual

3

viewports

desktop, tablet, and mobile

5

interaction types

menu, dialog, accordion,
tab, form focus

33

max variants

viewport and state matrix

Resource profiles

real loads styles, fonts, and images for
better contrast and layout accuracy.
minimal reduces cost but lowers accuracy.

Execution limits

Response size cap, navigation timeout,
axe timeout, page cap, scan cap, browser
health checks.

Static fallback

Limited HTML scan if Chromium fails. It is
low-confidence and not equivalent to
real-browser analysis.

URL security and SSRF protection

Opening customer-supplied URLs is one of the highest-risk parts of the system. Percevia treats URL validation as a layered security control.

1 Protocol and host

HTTP/S only, length limits, no credentials, no fragments, no reserved internal TLDs.

2 Network ranges

Private, loopback, link-local, multicast, IPv6 special ranges, and metadata IPs are blocked.

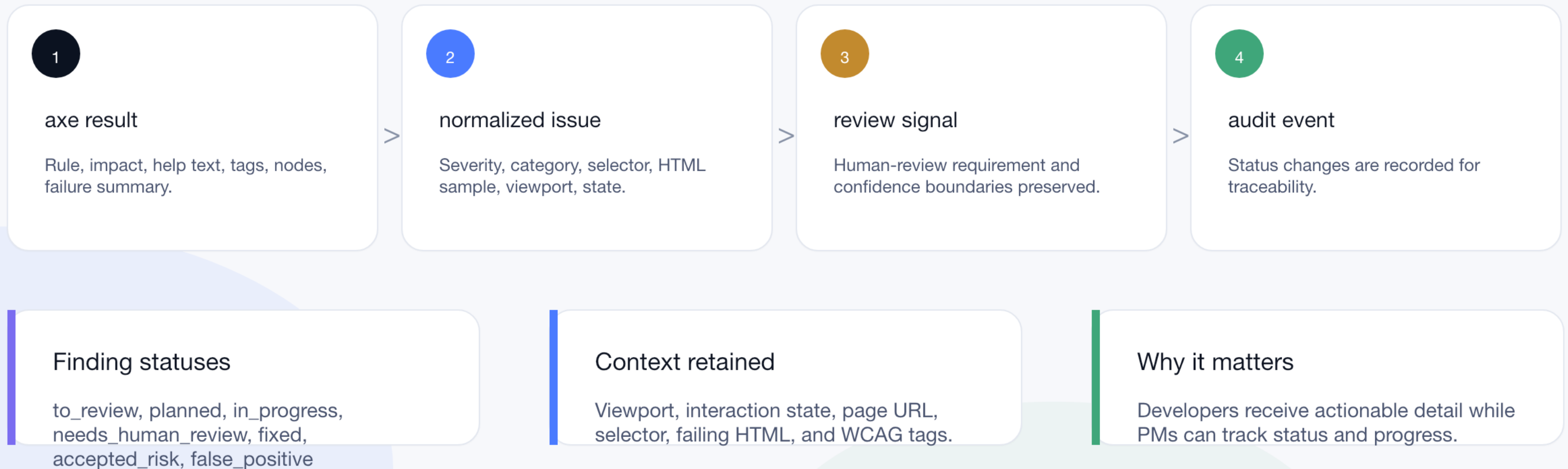
3 DNS and redirects

Every DNS result is validated, worker re-resolves, and final redirect URL is checked.

4 Scope control

Manual and sitemap scans enforce same-origin URLs and explicit authorization confirmation.

Finding normalization



Root-cause grouping

The system groups repeated defects into remediation targets so teams fix the underlying component rather than every repeated instance.

category + standard type + axe rule + normalized selector

Special case: contrast

Color-contrast issues can be grouped by foreground and background color pair.

Separation rules

Human-review findings, WCAG findings, and best-practice findings remain distinct.

Stored group data

Rule, title, worst severity, instance count, WCAG tag, remediation, and priority score.

Scoring and risk model

10

critical

highest impact
weight

6

serious

strong penalty

3

moderate

medium penalty

1

minor

low penalty

2

review

manual-review
weight

Occurrence factor

Repeated findings use a logarithmic multiplier with an upper bound, avoiding runaway scores.

Page normalization

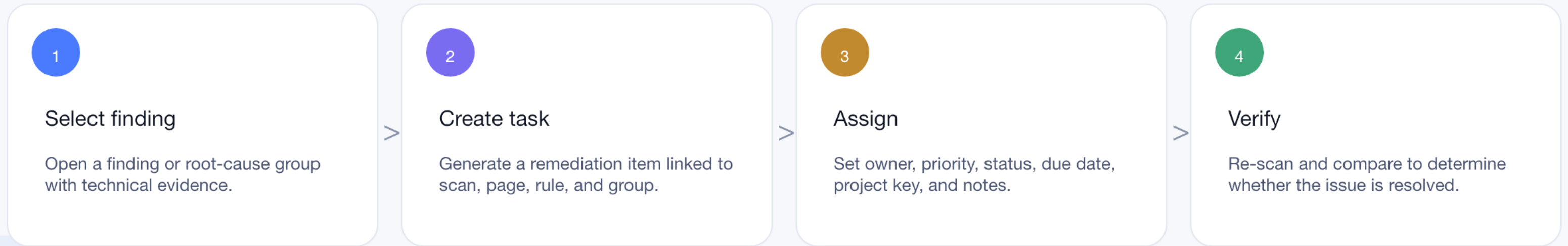
Total penalty is normalized with page count so larger scans are compared more fairly.

Outputs

0-100 score, A-F grade, risk level, page scores, category scores, WCAG count, best-practice count, review count.

Important: the score supports prioritization and trend analysis. It is not a legal compliance score.

Remediation management



Task data model

title, description, status, priority, assignee, due date, notes, axe rule, severity, source URL, scan and group relationships.

Project organization

Tasks can be organized by project or site domain, allowing one workspace to manage multiple websites or clients.

Reporting and public sharing

Report types

Full, executive, and CSV-oriented configurations.

Mandatory limits

Scope and legal-limitation notice cannot be removed.

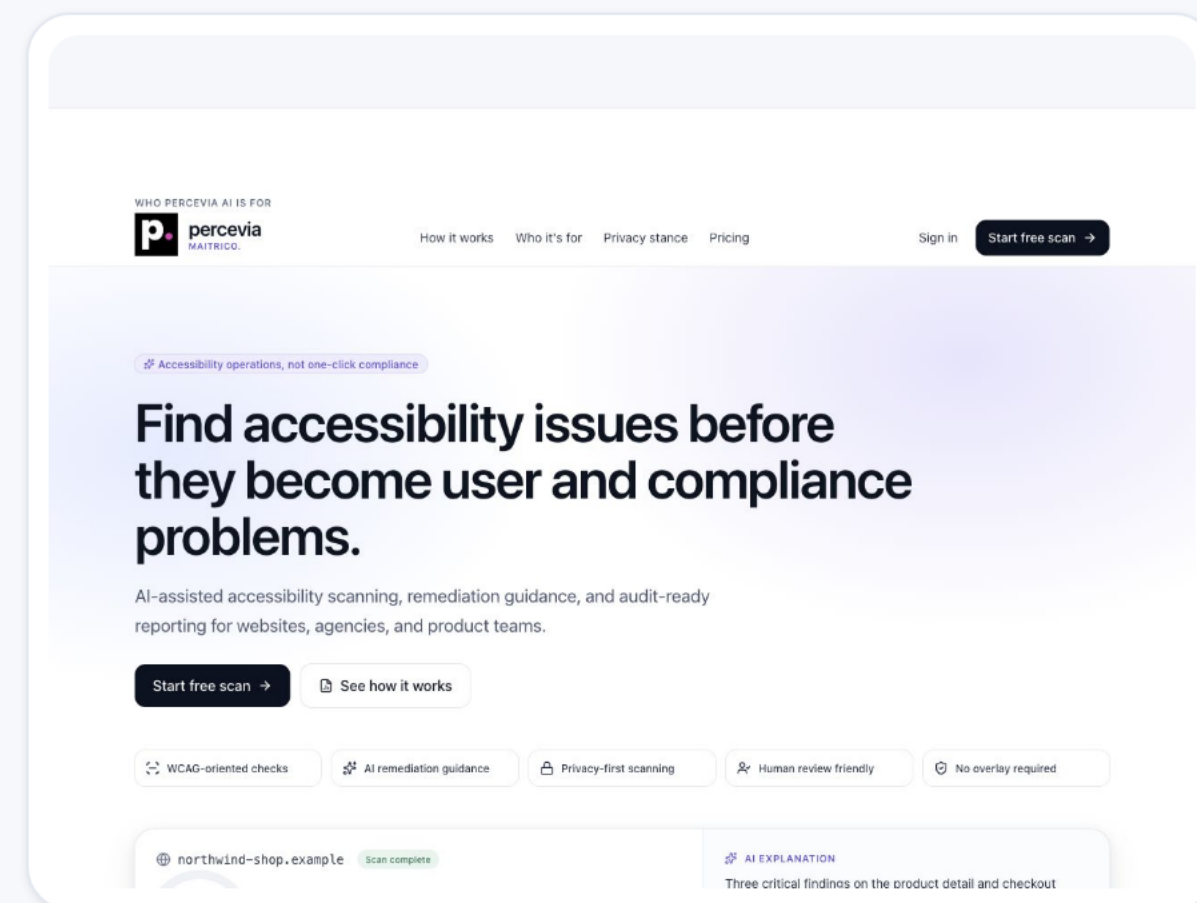
Export formats

HTML, PDF, CSV, and JSON. PDF depends on Chromium availability.

Public links

Random token, revocable, no-store, noindex, and gated by report export permission.

Agency, Team, and Enterprise plans can use workspace branding for shared reports.



AI layer

AI features are useful, but deliberately gated. The product treats AI as reviewable guidance, not an automatic compliance engine.

Use cases

Issue explanation, remediation summary, code example, framework-specific guidance, verification steps, client-friendly copy.

Privacy limits

Workspace consent required, HTML samples limited to 2 KB, no cookies, no form values, no screenshots.

Output safety

No certification claims, no legal guarantees, no 100% compliance claims, no overlay-as-substitute advice.

Default model: gpt-5.3-codex, configurable through environment variables. Every AI operation is rate-limited and written to audit logs.

Privacy and data management

Two-level screenshots

Workspace consent plus per-scan selection before visual evidence is captured.

Evidence controls

Selector, viewport, state, bounding box, redaction, expiry, and 650 KB persistence cap.

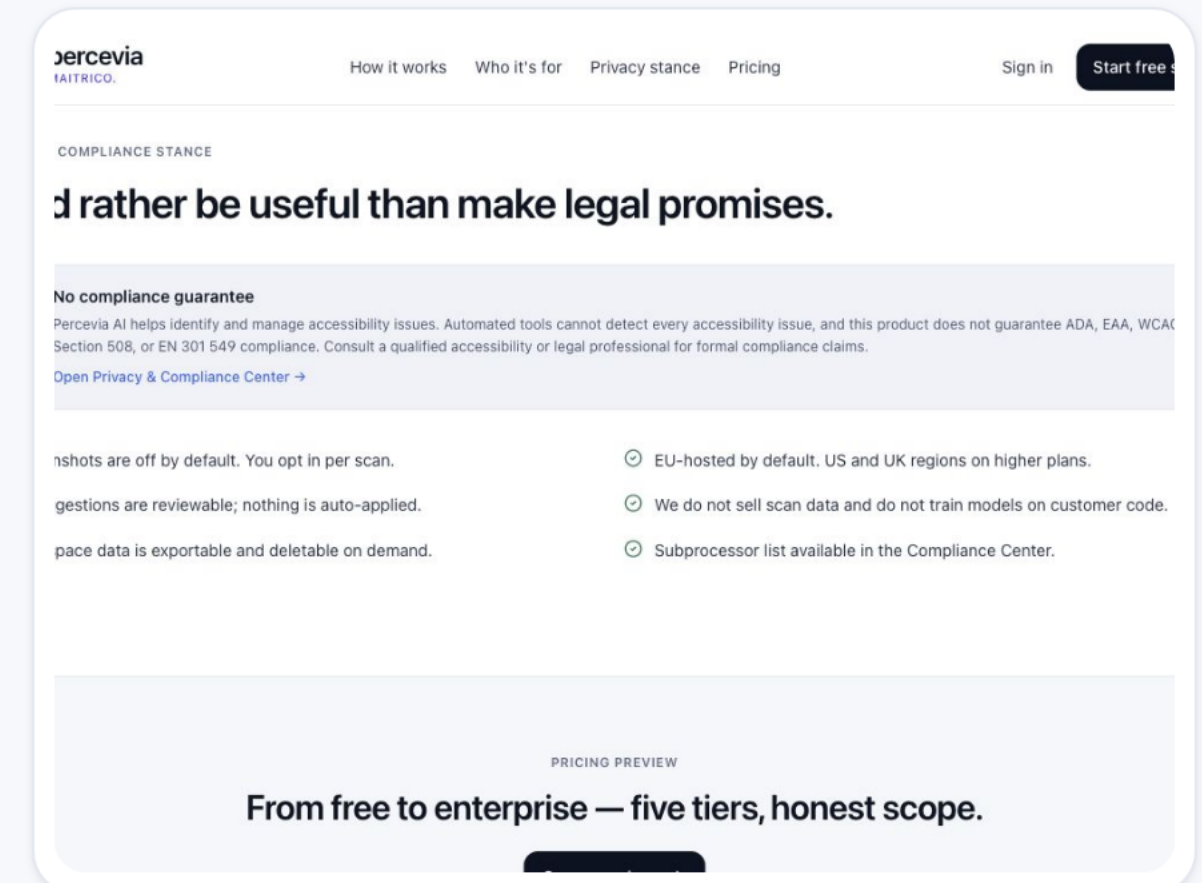
Data export

Workspace, privacy settings, scans, findings, tasks, reports, audit logs, and export timestamp.

Verified deletion

Async job deletes scan-related records and performs a final zero-residue verification pass.

V1 stores visual-evidence images as Base64 in Firestore. This simplifies infrastructure but is a known scaling constraint.



Continuous monitoring

Free	0	monitors	none
Starter	3	monitors	weekly
Agency	25	monitors	daily, every 3 days, weekly
Team	100	monitors	daily, every 3 days, weekly
Enterprise	500	monitors	daily, every 3 days, weekly

How it runs

Cloud Scheduler checks due monitors every five minutes by default. Due monitors reuse the standard scan pipeline.

Stored thresholds

New critical findings and score drops can be configured along with email and Slack channel settings.

Current gap

The delivery layer for actual email and Slack alerts is not implemented yet.

Subscription and entitlement model

Plans

Free, Starter, Agency, Team, and Enterprise.

Entitlements

Daily scans, pages per scan, evidence limits, seats, roles, monitors, white-label reports.

Polar integration

Checkout, customer portal, signed webhooks, active/trialing subscription grants paid access.

Production safety

Only verified Polar webhooks can assign paid plans in production.

Default scan limits range from Free 3 scans/day and 3 pages/scan to Enterprise 1000 scans/day and 1000 pages/scan.

The screenshot shows the Percevia pricing page with the following details:

- Navigation:** Back home, Sign in, Start free scan
- Privacy:** Privacy-first · No overlay
- Headline:** Pricing that scales with your team
- Subtext:** Honest tiers. No hidden "compliance" upsells. Percevia AI is positioned against one-click compliance — every plan reflects that.
- Plans:**
 - Free:** €0 forever. Try a single site. See whether Percevia AI fits your workflow.
 - ✓ 3 scans per day
 - ✓ Up to 3 pages per scan
 - ✓ 1 website
 - ✓ Basic findings report (web)
 - ✓ AI explanations (limited)
 - Starter:** €39 / month. Solo founders and freelancers managing a few sites.
 - ✓ 50 scans per day
 - ✓ Up to 50 pages per scan
 - ✓ 3 websites
 - ✓ AI explanations & remediation
 - ✓ PDF export
 - ✓ Email support
 - Agency (Most popular):** €129 / month. Agencies running audits across client sites.
 - ✓ 200 scans per day
 - ✓ Up to 200 pages per scan
 - ✓ Multiple client workspaces
 - ✓ Branded client reports
 - ✓ Remediation board
 - ✓ Priority support
 - Team:** €249 / month. Product teams collaborating on internal apps.
 - ✓ 500 scans per day
 - ✓ Up to 500 pages per scan
 - ✓ Roles & permissions
 - ✓ Advanced export (CSV / API)
 - ✓ Shared remediation backlog
 - ✓ SLA support
 - Enterprise Custom:** Regulated industries, large estates, and bespoke hosting.
 - ✓ Private / on-prem scanning (roadmap)
 - ✓ SSO (SAML / OIDC) — roadmap
 - ✓ Custom DPA
 - ✓ Dedicated region & residency
 - ✓ Local scanner roadmap

Core data model

1 Workspace boundary

Workspace, WorkspaceMember, UsageLimits, PrivacySettings

2 Scan domain

ScanJob, PageJob, ScanPage, AccessibilityIssue, IssueGroup, ScanSummary

3 Operations

VisualEvidence, RemediationTask, Report, Monitor, AuditLog

4 Global services

publicReportShares, invitations, deletion jobs, rateLimits, workerHeartbeats

Firestore workspace subcollections provide tenant separation; selected global collections handle sharing, invitations, deletion jobs, limits, and worker health.

Security posture

 HTTP-only Firebase session cookie	 Server-side membership checks	 Role and permission matrix	 Zod API validation
 SSRF and redirect validation	 Atomic Firestore job claims	 Private Cloud Run with OIDC	 Worker shared secret
 Scheduler bearer secret	 Polar signature verification	 Cryptographic share tokens	 Distributed rate limiting
 Audit logs	 AI data minimization	 Evidence redaction and expiry	 Encrypted local secrets

Rate limiting currently fails open if Firestore is unavailable. Platform-level abuse controls are recommended as a complementary layer.

Resilience and deployment

Atomic claims

Firestore transactions ensure that only one worker owns a scan or page job even when multiple workers compete.

Heartbeats and sweeper

15s worker heartbeat, 20s page-job heartbeat, stale recovery, timeout failure, and aggregation restart.

Chromium lifecycle

One browser process per worker, isolated contexts, concurrency 2 by default, memory recycling, crash relaunch.

Scale-to-zero

Cloud Run can idle to zero. Asynchronous queueing prevents cold starts from blocking user requests.

Deployment

Vercel web app, Cloud Run worker, Cloud Tasks queue, Cloud Scheduler sweeps and monitors.

Firestore ops

Composite indexes, TTL for rate limits, TTL for worker heartbeats, and operational runbooks.

Quality and test status

52

test files passed

1 skipped

323

tests passed

4 skipped

37.3k

TS/TSX lines

source and worker code

60

spec files

unit, integration, E2E

Passed

npm test, npm run typecheck, and npm run worker:typecheck all completed successfully.

Failed

npm run lint failed with 6 errors and 2 warnings, mainly hook state updates and explicit any usage.

Not run

Authenticated E2E tests and live production smoke testing were not run during this review.

Open risks and technical debt

P0

Accessibility statement security and claims

Stored XSS risk from unescaped fields; remove verified, conformant, and legal-protection language.

P1

Monitor alert delivery incomplete

Scheduler creates scans, but email and Slack alert delivery is not implemented.

P2

Firestore Base64 evidence

The 650 KB cap helps, but Cloud Storage is better for scale and cost.

P1

Lint quality gate failing

Typecheck and tests pass, but lint can block CI if enforced.

P1

PDF runtime dependency

Web runtime must launch Chromium; moving PDF rendering to the worker is safer.

P2

Region preference caveat

UI preference is not the same as physical data residency without deployment-level controls.

Recommended roadmap

Short term

- Escape statement fields
- Remove risky claims
- Fix lint failures
- Add monitor alert delivery
- Clean stale docs and labels
- Move PDF generation to worker

Medium term

- Move evidence to Cloud Storage
- Add monitor trends
- Email and Slack delivery
- Brand/language report templates
- Polar usage metrics
- Staging authenticated E2E CI

Long term

- Enterprise SSO
- Dedicated data regions
- Private scanner agent
- Large asset inventory
- SLA and regression analytics
- Jira, Linear, GitHub, GitLab integrations

The roadmap strengthens trust first, then scalability, then enterprise expansion.

What to emphasize in the presentation

1. Operational value

The core value is measurable, trackable, repeatable accessibility work, not simply finding more defects.

2. Trust posture

AI, screenshots, retention, public sharing, and legal language are governed explicitly.

3. Technical maturity

Async worker separation, SSRF defense, page jobs, retries, heartbeats, and tests show serious engineering.

4. Current honesty

The deck should clearly name the statement risk, lint gap, alert gap, and PDF runtime dependency.

5. Commercial path

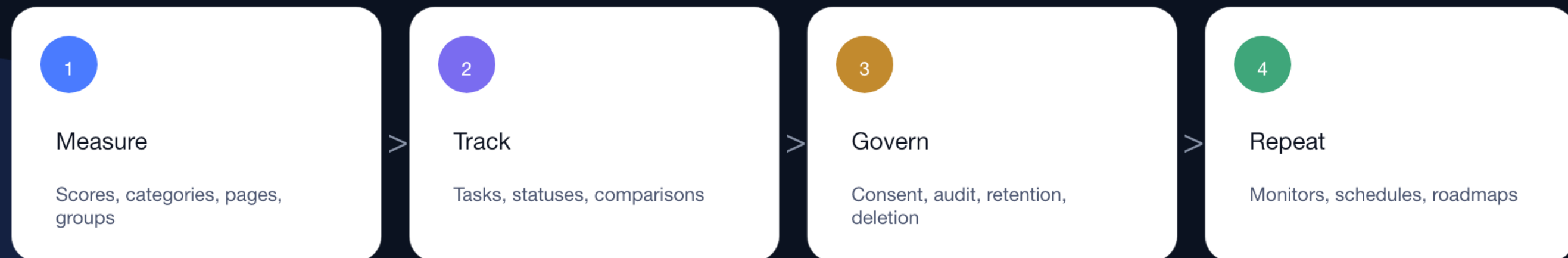
Polar subscriptions, team seats, monitor capacity, and report branding create a monetizable SaaS structure.

6. Future expansion

SSO, private scanning, data residency, inventory, trend analytics, and integrations map to enterprise buyers.

Percevia AI's value is not merely finding more defects.

It makes accessibility work measurable, trackable, governable, and repeatable.



Questions